# Difference in Differences

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ITAM

# Introduction

- Difference-in-Differences (DiD) is a quasi-experimental identification strategy to estimate causal effects.
- It uses natural variation in the implementation of treatment across different groups over time.

# John Snow and the Cholera Hypothesis

- John Snow used a natural experiment to show that cholera was transmitted through contaminated water.
- He compared cholera mortality rates between two water companies: Lambeth and Southwark and Vauxhall.
- Lambeth moved its water intake upstream, avoiding contamination, while Southwark and Vauxhall did not.

# Results from Snow's Study

- In 1849, Southwark and Vauxhall had 135 cholera cases per 10,000 households, while Lambeth had 85.
- In 1854, Southwark and Vauxhall had 147 cases per 10,000 households, while Lambeth had only 19.
- The estimated difference-in-differences (DiD) shows 78 fewer cholera deaths per 10,000 households in Lambeth compared to Southwark and Vauxhall.

# Estimating the Causal Effect

- The goal is to estimate the causal effect of clean water ($D$) on cholera deaths ($Y$).
- The basic DiD formula is:

$$\delta^{DD} = (Y_{\text{post, treated}} - Y_{\text{pre, treated}}) - (Y_{\text{post, control}} - Y_{\text{pre, control}}) \quad (1)$$

- This removes time fixed effects and time-invariant differences across groups.

Intuitively:

- $Y_{\text{post, treated}} - Y_{\text{pre, treated}} = \text{treatment effect} + \text{time trend}$
- $Y_{\text{post, control}} - Y_{\text{pre, control}} = \text{time trend}$
- $\delta^{DD} = \text{treatment effect}$

# Estimating the causal effect

$$\begin{aligned}\delta^{DD} &= (Y_{\text{post, treated}} - Y_{\text{pre, treated}}) - (Y_{\text{post, control}} - Y_{\text{pre, control}}) \\ &= (Y_{\text{post, treated}} - Y_{\text{post, control}}) - (Y_{\text{pre, treated}} - Y_{\text{pre, control}}) \qquad (2)\end{aligned}$$

Intuitively:

- $Y_{\text{post, treated}} - Y_{\text{post, control}} = \text{treatment effect} + \text{selection bias}$
- $Y_{\text{pre, treated}} - Y_{\text{pre, control}} = \text{selection bias}$
- $\delta^{DD} = \text{treatment effect}$

# DiD estimation

# 2×2

| | Treated | Control |
|---|---|---|
| Pre | $Y_{\text{pre, treated}}$ | $Y_{\text{pre, control}}$ |
| Post | $Y_{\text{post, treated}}$ | $Y_{\text{post, control}}$ |

- Intuitively, DiD is a comparison across the four cells.

# DiD estimation

- If $\delta$ is the true program impact, then:

$$\delta = \mathbb{E}[Y_i(1)|\, T_i = 1, t = \tau] - \mathbb{E}[Y_i(0)|\, T_i = 1, t = \tau] \tag{3}$$

- We assume $\delta$ does not depend on the time period, nor on individuals' characteristics.
- In the absence of the program, individual $i$'s outcome in period $t$ is given by:

$$\mathbb{E}[Y_i|\, T_i = 0, t = \tau] = \gamma_i + \lambda_\tau \tag{4}$$

- Selection bias is related to individuals' fixed characteristics $\gamma_i$.
- The time trend ($\lambda_\tau$) is the same for the treated and control groups: the **parallel trends assumption**.

# DiD estimation

Then:

$$
\begin{aligned}
\mathbb{E}[Y_{\text{pre, control}}] &= \mathbb{E}[Y_i(0)|T_i = 0, t = pre] = \mathbb{E}[\gamma_i|T_i = 0] + \mathbb{E}[\lambda_\tau|t = pre] \\
\mathbb{E}[Y_{\text{post, control}}] &= \mathbb{E}[Y_i(0)|T_i = 0, t = post] = \mathbb{E}[\gamma_i|T_i = 0] + \mathbb{E}[\lambda_\tau|t = post] \\
\mathbb{E}[Y_{\text{pre, treated}}] &= \mathbb{E}[Y_i(0)|T_i = 1, t = pre] = \mathbb{E}[\gamma_i|T_i = 1] + \mathbb{E}[\lambda_\tau|t = pre] \\
\mathbb{E}[Y_{\text{post, treated}}] &= \mathbb{E}[Y_i(1)|T_i = 1, t = pre] = \delta + \mathbb{E}[\gamma_i|T_i = 1] + \mathbb{E}[\lambda_\tau|t = pos
\end{aligned}
$$

# Comparisons

Treated vs. control:

$$\begin{aligned}\mathbb{E}[Y_{\text{post, treated}}] - \mathbb{E}[Y_{\text{post, control}}] &= \delta + \mathbb{E}[\gamma_i | T_i = 1] + \mathbb{E}[\lambda_\tau | t = post] - \\ &\quad \mathbb{E}[\gamma_i | T_i = 0] - \mathbb{E}[\lambda_\tau | t = post] \\ &= \delta + \underbrace{\mathbb{E}[\gamma_i | T_i = 1] - \mathbb{E}[\gamma_i | T_i = 0]}_{\text{selection bias}}\end{aligned}$$

Post vs. pre:

$$\begin{aligned}\mathbb{E}[Y_{\text{post, treated}}] - \mathbb{E}[Y_{\text{pre, treated}}] &= \delta + \mathbb{E}[\gamma_i | T_i = 1] + \mathbb{E}[\lambda_\tau | t = post] - \\ &\quad \mathbb{E}[Y_{i0} | T_i = 1, t = pre] \\ &\quad = \mathbb{E}[\gamma_i | T_i = 1] - \mathbb{E}[\lambda_\tau | t = pre] \\ &= \delta + \underbrace{\mathbb{E}[\lambda_\tau | t = post] - \mathbb{E}[\lambda_\tau | t = pre]}_{\text{time trend}}\end{aligned}$$

# Comparisons

$$
\begin{aligned}
\delta^{DD} &= (Y_{\text{post, treated}} - Y_{\text{pre, treated}}) - (Y_{\text{post, control}} - Y_{\text{pre, control}}) \\
&= (\delta + \mathbb{E}[\gamma_i | T_i = 1] + \mathbb{E}[\lambda_\tau | t = post] - \mathbb{E}[\gamma_i | T_i = 1] - \mathbb{E}[\lambda_\tau | t = pre]) - \\
&\quad (\mathbb{E}[\gamma_i | T_i = 0] + \mathbb{E}[\lambda_\tau | t = post] - \mathbb{E}[\gamma_i | T_i = 0] - \mathbb{E}[\lambda_\tau | t = pre]) \\
&= (\delta + \mathbb{E}[\lambda_\tau | t = post] - \mathbb{E}[\lambda_\tau | t = pre]) - \\
&\quad (\mathbb{E}[\lambda_\tau | t = post] - \mathbb{E}[\lambda_\tau | t = pre]) \\
&= \delta
\end{aligned}
$$

- DiD does not rely on homogeneous treatment effects.
- Under homogeneous treatment effects, the estimate corresponds to the average treatment effect on the treated (ATT).

# DiD as a regression

# DiD as a regression

We will estimate the following regression:

$$Y_{it} = \alpha + \beta T_i + \zeta Post_t + \delta \left( T_i * Post_t \right) + \varepsilon_{it}$$

where:

- $Post_t$ is a dummy equal to 1 if $t = 2$
- $\delta$ is the treatment effect.
- $\alpha = E[\gamma_i | T_i = 0] + \lambda_1$: pre-program mean for the control group.
- $\beta = E[\gamma_i | T_i = 1] - E[\gamma_i | T_i = 0]$: selection bias.
- $\zeta = \lambda_2 - \lambda_1$: time trend.

# DiD as a regression

Two-Way Fixed Effects (TWFE)

- With more than two time periods, statistical power increases.

$$Y_{i,t} = \alpha + \eta_i + \nu_t + \gamma D_{i,t} + \varepsilon_{i,t}$$

- $\eta_i$ are unit fixed effects.
- $\nu_t$ are time fixed effects.

# DiD as a regression

Event studies

- The idea is to include dummies for each post-treatment period.

$$Y_{i,t} = \alpha + \eta_i + \nu_t + \gamma_1 T1_{i,t} + \gamma_2 T2_{i,t} + \gamma_3 T3_{i,t} + \ldots + \varepsilon_{i,t}$$

When treatment intensity is continuous:

$$Y_{i,t} = \alpha + \beta Intensity_i + \zeta Post_t + \delta \left( Intensity_i * Post_t \right) + \varepsilon_{i,t}$$

# The parallel trends assumption

# The parallel trends assumption

- The key assumption is that outcomes in the treatment and control groups would follow the same time trend in the absence of treatment.
  - This does not imply they must have the same mean outcome level.
- However, the assumption cannot be directly tested.
  - Without it, we cannot identify the ATT.

# The parallel trends assumption

**How can we defend this assumption?**

1. Plot outcomes over time!
2. Falsification tests (look further into the past or into the future).
3. Control directly for time trends.

# Event study

- The regression we will estimate is:

$$Y_{its} = \gamma_s + \lambda_t + \sum_{\tau=-1}^{-q} \gamma_\tau T_{s\tau} + \sum_{\tau=0}^{m} \delta_\tau T_{s\tau} + x_{ist} + \varepsilon_{ist}$$

  - Treatment occurs at period zero.

# DiD with panel data

What exactly is $\beta^{DD}$?

$$Y_{it} = \alpha_i + \gamma_t + \beta^{DD} D_{it} + \varepsilon_{ti}$$

unit fixed effects

time fixed effects

treatment dummy

# DiD with panel data

**Frisch-Waugh (1933):** TWFE is equivalent to a univariate regression:

$$\tilde{Y}_{it} = \tilde{D}_{it} + \zeta_{ti}$$

where:

$$\tilde{Y}_{it} = Y_{it} - \bar{Y}_i - \left(\bar{Y}_t - \bar{\bar{Y}}\right)$$

and:

$$\tilde{D}_{it} = D_{it} - \bar{D}_i - \left(\bar{D}_t - \bar{\bar{D}}\right)$$

# Standard errors

# Standard errors

- Bertrand, Duflo, and Mullainathan (2004) point out that conventional standard errors often underestimate the standard deviation of the estimators. In other words, standard errors are biased downward.
- To address this, they propose:
  - Using bootstrap standard errors.
  - Using clustered standard errors at the group level.
  - Aggregating the data at the group level.